

Experiment Design Specification

Background & Motivation

The existing literature on gestures on at-issueness focuses mainly on referential gestures. Those are generally thought of as being not-at-issue (NAI). However, not much attention has been given to non-referential gestures – beats. The goal of this experiment is to investigate the effects beat gestures have on the (non-)at-issue status of expressions they accompany.

At-issueness has initially been thought of as a binary distinction. However, recent work has expanded these notions into a gradient way of thinking. In this experiment, I will focus on this gradient approach and see whether the presence of co-speech beat gestures increases the at-issue status of (on their own) non-at-issue constructions.

Research Questions

- Does a beat increase the gradient at-issue status of NAI constructions?
- Is there a difference in the amount of increase between various types of constructions?

Design

The data will be gathered in the form of an online experiment built using the Magpie framework. The participants will be presented with video recordings of a speaker. Adopting the ‘asking whether’ diagnostic from Tonhauser et al. 2018.

Materials

The set of stimuli consists of video clips which show a speaker producing an utterance. The speaker is captured from the waist up to the neck to avoid the influence of facial gestures and focus purely on hand gestures.

This experimental setup requires beat gestures to be produced at specific points during the utterance. The stimuli therefore could not be captured naturally. Therefore, the relevant beat gesture is the only gesture produced during the utterance. This of course could increase the effect of it since the ‘surprisal’/‘the amount to which it stands out’ is greater but the presence of and therefore further interaction with other gestures is beyond the scope of this paper.

During the utterance, the speaker in the videos holds their hands in a fixed resting position. both hands are in front of the body with palms facing inwards. The left hand is in front, and cupping the fingers of, the right hand. In the recordings, I take up the role of the speaker and since I am left-handed, my left hand is performing the beat. The thumb of the left hand is in this default position resting on the knuckle of the right hand and the index and middle

finger of the left hand rest on the knuckles of the ring and pinky finger. The beat gesture is then a simple flick of the wrist with the index and middle finger extended and the ring and pinky finger curled slightly towards the palm. The thumb is also lifted from its resting position but does not bend in either way. The beat is performed with a slight flick down and the fingers are spread so that it is not confusable with a pointing gesture.

Since what we want to observe is only the effect of the beat gesture, the same audio track is used in the video without the presence of a beat to ensure that the prosody does not change. This means that the focus might already be slightly increased since ‘the beat gestures presumably altered the prosody’

First, participants are shown a welcome screen and written instructions about the task.

Next, during the practice phase participants respond to two questions. A warning message will be shown if they don’t change the position of the slider or choose either 0 or 100 on the scale. The message will remind them that the goal is to grade on a scale. From each recording the participant will only see either a video with or without a beat.

In this experiment, we are focusing on 6 expressions with variable at-issueness as reported by Tonahuser 2018. Each expression is represented by two contexts and a control main clause sentence is included with further two contexts. Names were chosen randomly.

For each expression and context, there is a videoclip which either has or does not have a beat present. These two video clips are made from the same recording. The participant is presented each of the 14 stimuli once in a random order. The video clip plays twice each time. The participant sees for each expression one of the contexts with a beat and one without. This is done pseudorandomly by the ID of the participant in the system. In a bigger experiment, this could be done with a standard random function, but with a smaller pool of participants, I want to ensure a 50/50 split.

In increasing order of not-at-issue from Tonhauser et al. 2018

Testing phase

Does Lewis know_beat that Lucy brought her dog with her?

LC: Lucy brought her dog with her

Is Michael, a vegan_BEAT, going to bring his own food?

LC: Michael is a vegan

Main clause (control)

A. Is Phillip free_BEAT this weekend?

LC: Phillip is free this weekend

B. Did Mark get_BEAT his driving license?

LC: Mark got his driving license

C. Did Lily paint_BEAT the mug herself?

LC: Lily used to do ballet

D. Was Jim the youngest_BEAT in the class?

LC: Jim was the youngest in the class

Establish

A. Did Harley establish_BEAT that Phillip is free this weekend?

LC: Phillip is free this weekend

B. Did Amber establish_BEAT that Douglas brought enough food?

LC: Douglas brought enough food

Stop

A. Did Emily stop_BEAT coming to the meetings?

LC: Emily used to come to the meetings

B. Did David stop_BEAT talking to Jennifer?

LC: David was talking to Jennifer

Discover

A. Did Tiffany discover_BEAT that Marco stole a book from the library?

LC: Marco stole a book from the library

B. Did Christian discover_BEAT that the pie is not vegan?

LC: The pie is not vegan

Know

A. Does Linda know_BEAT that Olivia owns a farm in Utah?

LC: Olivia owns a farm in Utah

B. Does Lizzy know_BEAT that Martin moved away?

LC: Martin moved away

Nom appos

A. Is Jake, a student_BEAT, eligible for the discount?

LC: Jake is a student

B. Is Elsie, a dancer_BEAT, going to have issues with her joints?

LC: Elsie is a dancer

Annoyed

A. Is James annoyed_BEAT that Mark got his driving license?

LC: Mark got his driving license

B. Is Glen annoyed_BEAT that Casey found the book?

LC: Casey found the book

Adopt Tonhauser et al 2018 diagnostic ‘asking whether’.

A video shows the speaker asking a question. The participant then provides their answer to the question ‘Is the speaker asking whether LEXICAL_CONTENT?’ Using a scale with no and yes on each end. Answering closer to yes means the participant views the corresponding expression as not-as-issue to the extent it projects the lexical content.

Design & Variables

Independent variables

The experiment is a Yx2 factorial design with a between participants factor – presence or absence of a beat and a within participant factor which is the type of expression.

- Type of expression
- Beat - present or absent

Dependant variables

At-issue gradient

Measure response time?

Hypotheses

The presence of a beat gesture increases the AI status of NAI expression.